

Research Directions on Hybrid PON for Very High Speed PON

Ivan N. Cano⁽¹⁾

⁽¹⁾ Huawei Heisenberg Research Center, Riesstr. 25, 80992 Munich, Germany ivan.cano@huawei.com

Abstract Two systems are proposed for VHSP that aim to centralize the complexity at the OLT while having simple DD at the ONU-Rx: DCPC and SSB. An evaluation of both systems in terms of their CD tolerance is provided as well as motivations for further research. ©2026 The Author(s)

Introduction

Following the publication of the latest ITU-T recommendation for 50 Gb/s passive optical networks (PON) in 2021 [1], the ITU-T initiated an exploratory study for the next PON generation, designated as very high speed PON (VHSP) [2]. Concurrently, the Full-Service Access Network (FSAN) initiative published a roadmap addressing anticipated operator needs for VHSP [3]. The primary requirements for this future system include delivering a service capacity of at least 200Gb/s, enabling service convergence, and supporting coexistence of multiple PON generations within existing optical distribution networks (ODN) [2]. To align these requirements with potential technological solutions, the ITU-T recently agreed upon the first version of supplement G.Suppl.88 [4], which documents system requirements and evaluates potential technology candidates to support them.

Two immediate technological options are under consideration: intensity-modulation with direct-detection (IM-DD) and coherent detection (CO). IM-DD is a natural choice, having been employed in all previous PON generations, it benefits from an established ecosystem. Through the use of DSP and optical amplification, various studies have demonstrated the feasibility of IM-DD for VHSP in achieving the required bitrate and power budgets. However, its limited chromatic dispersion (CD) tolerance restricts the operating wavelengths to the vicinity of 1310 nm [2,5,6], a band already occupied by GPON upstream (US). Conversely, CO offers advantages such as enhanced receiver (Rx) sensitivity and high tolerance to CD and polarization mode dispersion (PMD), enabling operation at any wavelength and ensuring coexistence with legacy PON technologies. Nevertheless, CO faces challenges in the PON context. Although cost reductions in CO transceivers driven by data-centre applications [7] are expected, it remains questionable whether they can be translated directly to PON, as modules must support bidirectional operation over a single fibre and burst-mode reception in the US requires fast DSP convergence and laser alignment [8-10]. These challenges are currently under active study, and a set of recommendations for a 100 Gb/s CO based PON has recently been published [10].

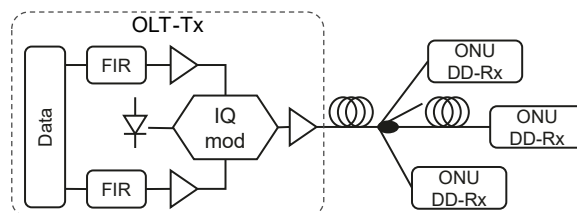


Fig. 1: General hybrid architecture schematic

This paper describes an alternative system with the aim to bridge the gap between IM-DD and CO by leveraging components from both technologies across different parts of the network. This approach is currently categorized under the section 'alternative direct detection schemes' in G.Suppl.88. Specifically, the complexity of a CO transmitter is adopted at the optical line terminal (OLT) while maintaining the simplicity of DD at the optical network unit (ONU). This strategy centralizes complexity at the OLT, thereby maintaining a simple ONU implementation. To satisfy coexistence requirements, either digital chromatic dispersion compensation (DCPC) or single side-band (SSB) modulation is implemented at the OLT to mitigate the CD induced penalties. While both methods can be realized in either the digital or optical domain, this paper focuses on their digital implementation. DCPC and SSB have been studied for PON in [11] and [12] respectively and this paper summarizes key results regarding these techniques with the aim of motivating further research.

General architecture

Fig. 1 illustrates the general architecture for the proposed hybrid solution. Since the OLT is shared across the PON, it can accommodate optical transceivers of higher complexity than the cost-sensitive ONUs. Consequently, the ONU Rx should preferably utilize DD. To achieve the Rx sensitivities required for PON, PAM2 modulation is employed, with an optically pre-amplified Rx based on a semiconductor optical amplifier (SOA) including optical filtering and DSP after photodetection (as introduced in 50G-PON which also utilizes PAM2). The DSP performs equalization to compensate for any residual inter-symbol interference (ISI) caused by CD or limited component bandwidth (BW).

Wavelength overlay is anticipated to achieve multi-generation coexistence and, as seen in Fig. 2, the spectral occupancy of current PON generations leaves very few regions available for VHSP. Following the idea of having an ONU based on DD, it is beneficial to operate in O-Band where the CD is low. However, for the expected bitrates of at least 120 Gb/s/λ (considering the use of two wavelengths to achieve the net bitrate of 200 Gb/s), CD can induce severe ISI even at O-Band wavelengths. To enhance CD tolerance, two methods are proposed for the OLT: DCPC and SSB modulation. While both approaches require an optical IQ-modulator (IQM), thereby adding complexity, this component is located at the OLT, centralizing the cost and complexity.

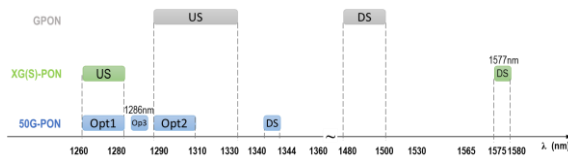


Fig. 2: Wavelength bands of current PON systems.

Digital CD Pre-Compensation

DCPC was originally proposed in [13] for CD limited fibre links and later evaluated by the same group for PON in [11]. The principle involves pre-distorting the signal to counteract a specific amount of CD, ensuring the signal appears undistorted after transmission. This is achieved digitally by using a complex-valued FIR filter with coefficients calculated according to [14]. While the method is effective for point-to-point links where the accumulated CD is fixed, PONs feature ONUs that can be distributed along the whole ODN. Hence, using DCPC directly implies that some ONUs would receive a clean signal while others detect a distorted one. The residual CD is subsequently compensated at the ONU via DSP to reduce the penalty and increase the tolerance.

Fig. 3 shows the optical path penalty (OPP) obtained through simulations of a conventional

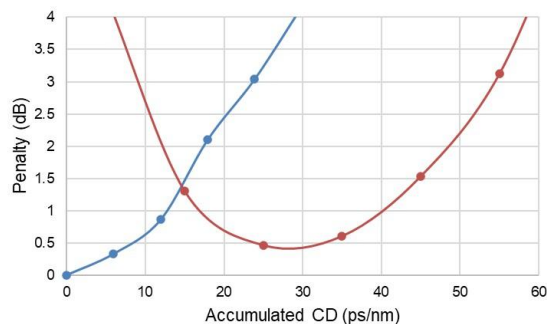


Fig. 3: Penalty versus accumulated CD for conventional PAM2 and PAM2-DCPC pre-compensated 30ps/nm

120 GBaud PAM2 signal and the penalty of a DCPC signal (pre-compensated for 30 ps/nm) with reference to the PAM2 BtB OMA Rx sensitivity. These curves were computed at a pre-FEC BER of $2 \cdot 10^{-2}$ using a 23-tap FFE and 3-bit memory MLSE at Rx. The latter consisted of an SOA (gain of 20 dB and NF of 6 dB), followed by a 4 nm optical filter and detected with a p-i-n PD with 65 GHz BW. For the conventional PAM2 case, the maximum accumulated CD at a 3 dB penalty is ~ 25 ps/nm. Although DCPC incurs in a BtB penalty due to signal pre-distortion, this penalty vanishes at 30 ps/nm as the fibre CD compensates the optical signal, ultimately extending the accumulated CD to ~ 55 ps/nm (at 3 dB penalty limit) and doubling the CD tolerance of conventional PAM2. Translating this to the worst-case CD of G.652 SMF after 20 km results in a maximum possible operating wavelength of 1331 nm.

However, to achieve coexistence with previous PON technologies, higher operating wavelengths would be needed and a possible way would be to group the ONUs according to their accumulated CD. Then, multiple sets of FIR coefficients could be stored to pre-compensate for different CD values across different time-slots [11]. Additionally, these coefficients could be further optimized using metrics from the US signals, as proposed in [15]. The primary advantage of digital pre-distortion is that different CD values can be addressed by simply changing the coefficients through software; however, introducing ONU groups adds complexity to the management protocols. To minimize this, operation could be limited to the 1360–1370 nm range, which allows coexistence with previous PON generations while restricting the required number of groups [16].

Since certain ONUs may intermittently fail the BER threshold when their group is not being transmitted, a mechanism is required to prevent loss of control data or clock synchronization. A common header sent at a lower bitrate, as proposed in [17], could be detected by all ONUs to ensure robustness. In addition, to achieve the expected bitrate of 200 Gb/s, a possibility is to use two wavelengths at 120 Gb/s each (accounting for overhead). Then one wavelength could carry control and clock synchronization for all ONUs alongside data, while the second would be dedicated solely to data transmission.

With an optical IQM already present at the OLT Tx, optical duobinary (ODB) modulation could also be employed. This format requires a differential pre-coder and a delay-and-add function which can be implemented using a low-pass filter with a specific BW. Additionally, the electrical driver must generate a signal with sufficient amplitude to accommodate the $0 - \pi$ shifts in the

IQM. The FIR filter coefficients are computed identically to those for NRZ, and the Rx configuration remains unchanged. Fig. 3 shows the OPP for the ODB signal obtained through simulations, both without DCPC and with 90 ps/nm pre-compensation. ODB exhibits superior CD tolerance compared to NRZ, achieving 40 ps/nm at a 3 dB penalty. As observed with NRZ, DCPC introduces a BtB penalty that disappears at the compensated CD value, subsequently extending the attainable accumulated CD to 120 ps/nm. Translating this to the worst-case CD over 20 km yields a maximum possible operating wavelength of 1370 nm. Thus, with a couple of groups an accumulated CD of up to 120 ps/nm could be covered, ensuring coexistence with previous PON technologies. We anticipate that further optimization of the system parameters could enhance performance, thereby simplifying the PON operation and management.

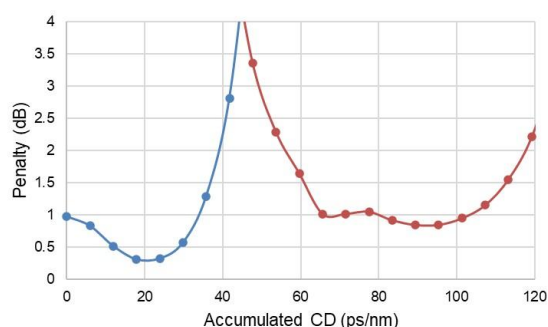


Fig. 4: Penalty referenced to PAM2 BtB versus accumulated CD for ODB modulation and ODB-DCPC with 90 ps/nm CD pre-compensation.

Single Sideband Modulation

SSB modulation reduces CD-induced penalties by mitigating the destructive interference between the upper and lower sidebands of an optical signal [18]. While several methods exist to generate an optical SSB signal, this section focuses on a digital implementation employing a FIR filter to produce a Hilbert transform before modulation through an IQM [18]. To employ DD at the ONU Rx, the optical carrier must be

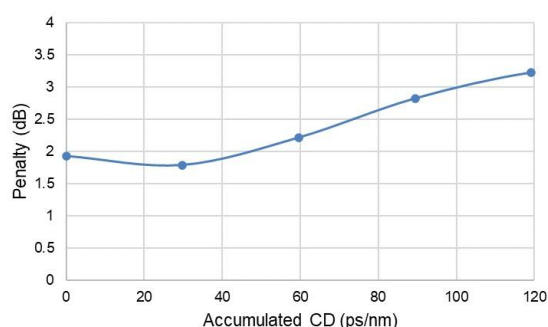


Fig. 5: Penalty referenced to PAM2 BtB versus accumulated CD for SSB modulation

preserved; furthermore, the carrier to signal power ratio (CSPR) should be optimized for best performance. The SSB penalty obtained from simulations is displayed in Fig. 4, using the PAM2 BtB results as reference. The optical Rx structure and DSP is identical to DCPC case, with the Hilbert transform generated with an asymmetrical 15-taps FIR filter. As shown, the CD tolerance improved, with a penalty of only 3.2 dB relative to PAM2 BtB after 120 ps/nm of accumulated CD. When referenced to SSB BtB, this penalty decreases to 1.2 dB. This would provide sufficient CD tolerance to avoid ONU grouping, thereby simplifying the PON management. Given this high CD tolerance, the bitrate per-wavelength could be scaled, provided that high-BW components currently under research [19-21] become mature. Alternatively, the Rx DSP could be optimized while maintaining a penalty below the 3 dB margin. In addition, there is a variety of SSB implementation schemes which could also be feasible solutions.

Conclusions

This paper presented DCPC and SSB techniques to enhance the CD tolerance for VHSP DS while keeping ONU receivers based on DD. Both approaches centralize the complexity at the OLT, where a single polarization IQM is driven by digital complex-valued FIR filters to either pre-compensate CD or suppress a spectral sideband. While both techniques significantly improve CD tolerance, achieving full coexistence with legacy PON systems using DCPC requires grouping the ONUs according to their location and accepting a residual penalty to avoid individual pre-compensation. In contrast, SSB provides equivalent resilience without grouping, offering an advantage for network management.

Even if both DCPC and SSB require a more complex OLT Tx than IM, they demonstrate enhanced CD tolerance while keeping DD at the ONU Rx. This study provides a foundational analysis focused primarily on the CD impairment; however the impact of other transmission effects like PMD and nonlinearities should also be considered as metrics to evaluate the most efficient modulation scheme.

Although this paper explored digital implementations, optical approaches could be equally viable. Furthermore, the US transmission remains challenging and solutions are needed for its unique demands, e.g. burst-mode operation while keeping the premise of having an ONU transmitter as simple as possible. Future research in this direction is expected together with system optimization to fully realize a complete VHSP solution that fulfils all requirements while maintaining implementation simplicity.

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